

Exercise 2 – Employee Salary Report

```
airflow@137ba4301055:/opt/airflow/dags$ cd /opt/airflow/dags
airflow@137ba4301055:/opt/airflow/dags$ cat > exercise2.py
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime
```

```
def create_salary_file():
    with open('/tmp/employees.txt', 'w') as f:
        f.write(
            "Rahul,45000\n"
            "Priya,52000\n"
            "Amit,61000\n"
            "Sneha,48000"
        )

def calculate_total_salary():
    total_salary = 0

    with open('/tmp/employees.txt', 'r') as f:
        for line in f:
            name, salary = line.strip().split(',')
            total_salary += int(salary)

    with open('/tmp/total_salary.txt', 'w') as f:
        f.write(str(total_salary))

def generate_report():
    with open('/tmp/total_salary.txt', 'r') as f:
        total_salary = f.read()

    with open('/tmp/employees.txt', 'r') as f:
        employee_count = len(f.readlines())
```

```
airflow@137ba4301055:/opt/airflow/dags$ airflow dags reserialize
2026-06-23T09:48:37.787203Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:37.788363Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:37.789156Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:37.790124Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:37.790880Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:37.791625Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:38.479039Z [info] ] DAG bundles loaded: dags-folder [airflow.dag_processing.bundles.manager.DagBundlesManager] loc=manager.py:209
2026-06-23T09:48:38.481464Z [info] ] Filling up the DagBag from /opt/airflow/dags [airflow.dag_processing.bundles.manager.DagBundlesManager] loc=manager.py:210
```

```
airflow@137ba4301055:/opt/airflow/dags$ airflow dags list
2026-06-23T09:48:52.920933Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:52.921783Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:52.922718Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:52.923845Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:52.924833Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T09:48:52.925975Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
dag_id | fileloc | owners | is_paused | bundle_name | bundle_version
-----+-----+-----+-----+-----+-----
exercise1_file_read | /opt/airflow/dags/exercise1.py | airflow | False | dags-folder | None
exercise2_salary_report | /opt/airflow/dags/exercise2.py | airflow | True | dags-folder | None
```

TASK 3

The screenshot shows the Apache Airflow interface with the 'Dags' tab selected. Two DAGs are listed:

- exercise1_file_read**: Latest Run on 2026-06-23 15:01:16, Status: Success.
- exercise2_salary_report**: Latest Run on 2026-06-23 15:22:36, Status: Success.

The screenshot shows the 'Dag Run' view for 'exercise2_salary_report'. The run is manual, dated 2026-06-23T09:52:36.915301+00:00, and is in a 'Success' state. The task instances table shows the following details:

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
generate_report		Success	2026-06-23 15:22:39	1	PythonOperator	00:00:00.460
calculate_total_salary		Success	2026-06-23 15:22:38	1	PythonOperator	00:00:00.189
create_salary_file		Success	2026-06-23 15:22:37	1	PythonOperator	00:00:00.304

OUTPUT

```
airflow@137ba4301055:/opt/airflow/dags$ cat /tmp/report.txt
Salary Report
Employees = 4
Total Salary = 206000airflow@137ba4301055:/opt/airflow/dags$ cat /tmp/employees.txt
Rahul,45000
Priya,52000
Amit,61000
Sneha,48000airflow@137ba4301055:/opt/airflow/dags$ cat /tmp/total_salary.txt
206000airflow@137ba4301055:/opt/airflow/dags$
```